



AI-Driven Protein Structure Prediction: The AlphaFold Revolution in Structural Biology



Solving a 50-year-old protein folding problem using artificial intelligence

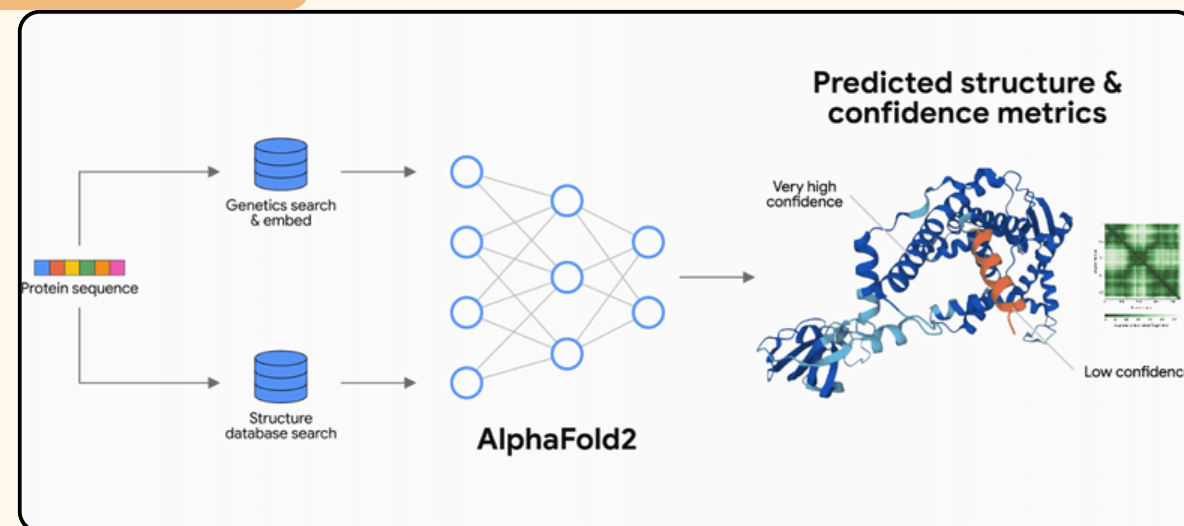
INTRODUCTION

- Proteins perform essential biological functions, and their activity depends on their 3D structure.
- Predicting protein structure from amino acid sequence is known as the protein folding problem.
- Traditional methods like X-ray crystallography and NMR are accurate but slow and expensive.
- AI-based tools like AlphaFold can predict protein structures rapidly and with high accuracy.

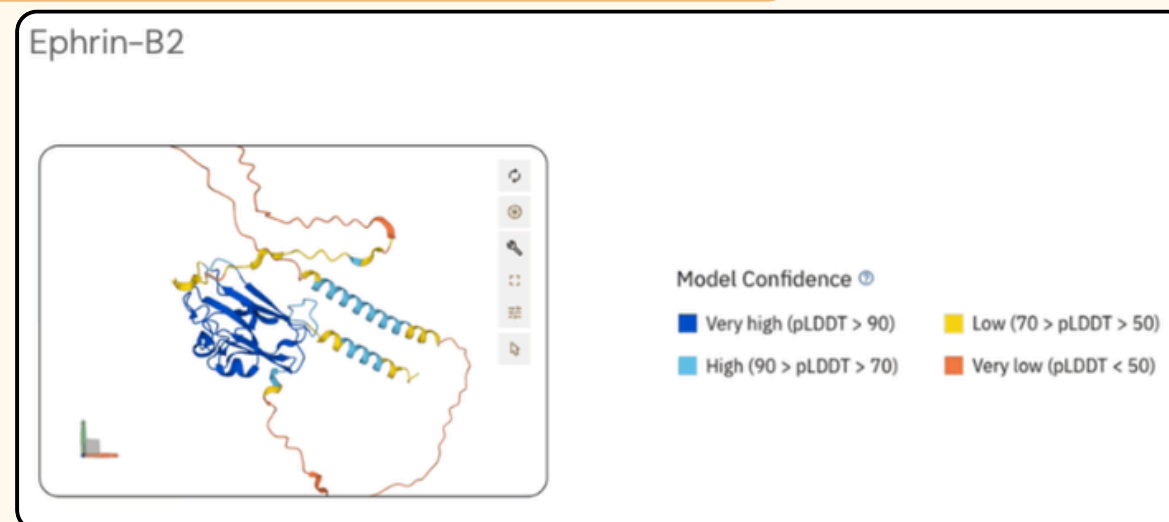
PROBLEM STATEMENT

- Protein structure determination is essential for understanding biological function.
- Experimental techniques are:
 - Time-consuming
 - Expensive
 - Not feasible for all proteins
- Many proteins (especially membrane proteins) are difficult to study experimentally
- Need for faster, scalable, and reliable prediction methods

METHODOLOGY



pLDDT: Understanding local confidence



pLDDT score indicates confidence in predicted protein structure

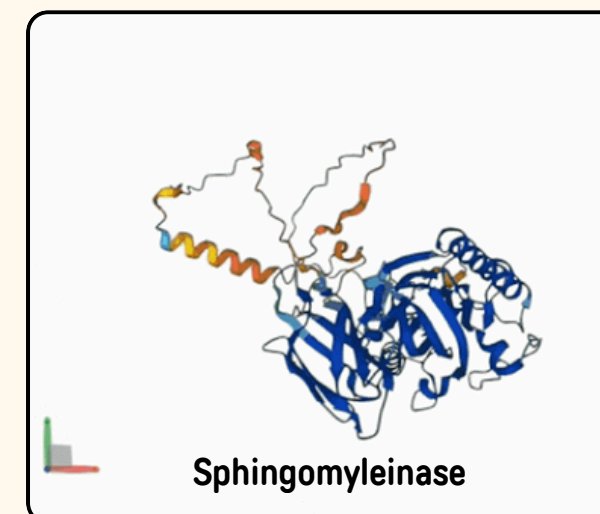
KEY FEATURES

- High accuracy (near experimental level)
- Fast prediction (minutes to hours)
- No need for crystallization
- Scalable for millions of proteins

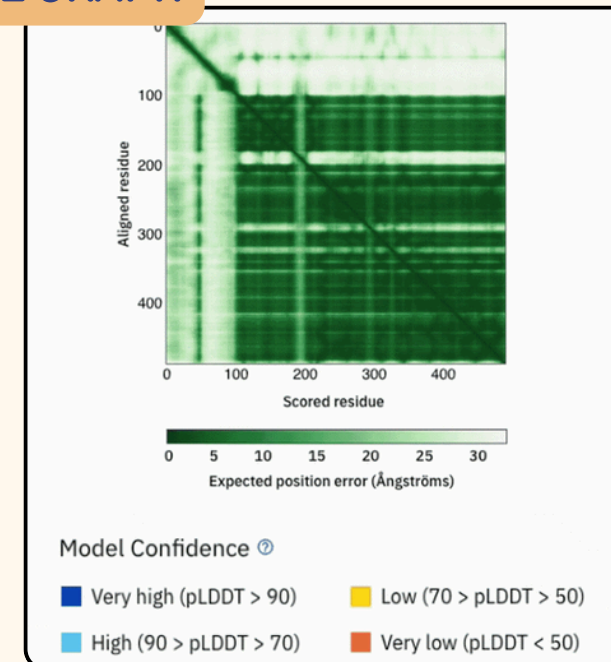
APPLICATIONS

- Drug discovery
- Disease research (Alzheimer's, cancer)
- Protein engineering
- Understanding biological pathways

PAE (Predicted aligned error)



PAE GRAPH



RESULTS

- Solved a 50-year scientific challenge
- Millions of protein structures predicted
- Widely used in research worldwide
- Accelerated biological discoveries
- Predicted 200+ million protein structures

LIMITATIONS

- Less accurate for:
 - protein complexes
 - dynamic structures
- Does not fully replace experimental methods
- Predictions require validation

CONCLUSION

- AlphaFold represents a major breakthrough in computational biology
- Enables rapid and accurate protein structure prediction
- Acts as a powerful complement to experimental techniques
- Future advancements will improve prediction of interactions and dynamics

REFERENCES

- AlphaFold Database (EMBL-EBI)
- Jumper et al., 2021 (Nature)
- AlphaFold training (EBI)
- CASP14 competition results